Conjecture 20 is false

A counterexample at $p = 3$, weight decay, $n = 11$, with proof

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Abstract

Conjecture 20 of *Feature Sparsity via Margin Maximization* asserts that for every odd prime p, every $n \geq 1$ and every $2 \leq b \leq \nu = n + 1$, the maximum $L_{2,b}$-margin over all product networks equals the interference-free value $\gamma^\star_{b,\mathrm{IF}} = \nu^{1-s} B_p \Gamma_s(p)$ of Theorem 8 ($s = \nu/b$). We give an explicit product network for $p = 3$, $n = 11$, $b = 2$ of width 8 whose normalized margin exceeds $\gamma^\star_{b,\mathrm{IF}}$ by a factor 1.0544, and prove this by a finite computation in exact rational arithmetic, which reduces to the one-line inequality $2.7312 > 3(P/72)^6/12^5 = 2.5904$. We also prove failures at $n = 13$ (factor 1.109, exact), $n = 14$ (factor 1.429), and on whole intervals of exponents, $b \in [2, 2.13]$ at $n = 12$ (interval arithmetic). Finally, we show why the two earlier proof strategies could not have reached the critical range: the relaxation they work in, which keeps only the clean and sign-flipped Fourier modes, already violates the conjectured bound at $n = 3$.

1 Setting

We use the notation of the paper. For $x = (x_1, \ldots, x_n) \in \mathbb{Z}_p^n$ let $y(x) = x_1 + \cdots + x_n \bmod p$. A *product network* of width m is

$$f(\theta, x)[c] = \sum_{i=1}^{m} w_i(c) \prod_{t=1}^{n} u_{it}(x_t), \qquad \omega_i = (u_{i1}, \ldots, u_{in}, w_i) \in (\mathbb{R}^p)^{n+1},$$

with homogeneity $\nu = n + 1$. Write $\|\theta\|_{2,b} = \big(\sum_i \|\omega_i\|_2^b\big)^{1/b}$, where $\|\omega_i\|_2^2 = \sum_t \|u_{it}\|_2^2 + \|w_i\|_2^2$. The margin is

$$h(\theta) = \min_{x \in \mathbb{Z}_p^n} \min_{c \neq y(x)} \big(f(\theta, x)[y(x)] - f(\theta, x)[c]\big),$$

and the normalized margin is $h(\theta)/\|\theta\|_{2,b}^\nu$. Theorem 8 of the paper gives the interference-free optimum

$$\gamma^\star_{b,\mathrm{IF}} = \nu^{1-s} B_p \Gamma_s(p), \qquad s = \nu/b, \qquad B_p = 2(2p\nu)^{-\nu/2},$$

attained at every width $m \geq \nu |\mathrm{spt}(y^\star)|$. Conjecture 20 states that no product network of any width has normalized margin larger than $\gamma^\star_{b,\mathrm{IF}}$.

Lemma 1 (the conjectured value at $p = 3$)**.** *For $p = 3$ and every $s \geq 1$, $\Gamma_s(3) = 3/2$. Hence*

$$\gamma^\star_{b,\mathrm{IF}} = \tfrac{3}{2}\, \nu^{1-\nu/b} \cdot 2(6\nu)^{-\nu/2}, \qquad \textit{and at } b = 2: \quad \gamma^\star_{2,\mathrm{IF}} = 3\nu\,(6\nu^2)^{-\nu/2}.$$

In particular $\gamma^\star_{2,\mathrm{IF}} = 3/(12^5 \cdot 72^6)$ for $n = 11$ and $\gamma^\star_{2,\mathrm{IF}} = 42 \cdot 1176^{-7}$ for $n = 13$.

Proof. For $p = 3$ the set of frequencies is $F = \{1\}$ and $\kappa_1(\delta) = 1 - \cos(2\pi\delta/3) = 3/2$ for $\delta \in \{1, 2\}$. Program (2) of the paper is therefore $\max\{\frac{3}{2}y_1 : y_1 \geq 0,\ y_1^{1/s} \leq 1\} = \frac{3}{2}$. Substitute into $\nu^{1-s}B_3\Gamma_s(3)$. At $b = 2$, $s = \nu/2$ and $\nu^{1-\nu/2} \cdot 3(6\nu)^{-\nu/2} = 3\nu(6\nu^2)^{-\nu/2}$. For $\nu = 12$: $36 \cdot 864^{-6} = 3/(12^5 \cdot 72^6)$. For $\nu = 14$: $42 \cdot 1176^{-7}$. □

Lemma 2 (slot-symmetric networks)**.** *Suppose $u_{it} = u_i$ for all slots t. For $x \in \mathbb{Z}_3^n$ let $N_a(x) = \#\{t : x_t = a\}$. Then*

$$f(\theta, x)[c] = \sum_i w_i(c)\, u_i(0)^{N_0} u_i(1)^{N_1} u_i(2)^{N_2}, \qquad y(x) \equiv N_1 + 2N_2 \pmod 3.$$

Hence $h(\theta)$ is the minimum, over the $\binom{n+2}{2}$ vectors $N = (N_0, N_1, N_2) \in \mathbb{Z}^3_{\geq 0}$ with $N_0+N_1+N_2 = n$, of $\min_{c \neq y}\big(f_N[y] - f_N[c]\big)$, where f_N is the common logit vector of all inputs with counts N.

Proof. The product $\prod_t u_i(x_t)$ is invariant under permuting the coordinates of x, so it depends only on the multiset of values, that is on $N(x)$. Also $\sum_t x_t = 0 \cdot N_0 + 1 \cdot N_1 + 2 \cdot N_2$. Every N with $|N| = n$ is attained by some x, so minimizing over x and over N gives the same value. □

2 The counterexample

Theorem 3. *Let $p = 3$, $n = 11$ (so $\nu = 12$) and $b = 2$. Let $\theta^\star$ be the width-8 product network with $u_{it} = u_i$ for all $t = 1, \ldots, 11$ and the weights of Table 1. Then*

$$\frac{h(\theta^\star)}{\|\theta^\star\|_{2,2}^{12}} \;>\; 1.054 \cdot \gamma^\star_{2,\mathrm{IF}}.$$

Consequently, for every width $m \geq 8$ (padding with zero neurons), and in particular for every width $m \geq 12$ at which Theorem 8's value is attained, the maximum $L_{2,2}$-margin over product networks is strictly larger than the interference-free value. Conjecture 20 is false.

i	$u_i(0)$	$u_i(1)$	$u_i(2)$	$w_i(0)$	$w_i(1)$	$w_i(2)$
1	−0.4846467	0.0986351	2.3735957	0.9287971	1.0496311	−1.9784282
2	−0.9492087	0.4121459	2.1787251	−0.7213797	−1.2263075	1.9476873
3	−1.2303519	0.7873084	1.9158950	0.4609319	1.4256721	−1.8866040
4	−1.3685236	1.1830027	1.5772570	−0.1610123	−1.6108253	1.7718376
5	−1.3684776	1.5721532	1.1868510	−0.1574121	1.7692451	−1.6118330
6	−1.2303770	1.9114843	0.7892217	0.4583909	−1.8837781	1.4253872
7	−0.9512944	2.1719362	0.4152123	−0.7173870	1.9434179	−1.2260309
8	−0.4871743	2.3639747	0.1000316	0.9243126	−1.9711611	1.0468485

Table 1: The network $\theta^\star$ ($p = 3$, $n = 11$, width 8). Every input slot of neuron i uses the factor u_i. All entries are exact 7-decimal rationals.

Proof. All weights are rational, so every quantity below is a rational number computed exactly.

Step 1 (the norm). By definition $\|\omega_i\|_2^2 = 11\|u_i\|_2^2 + \|w_i\|_2^2$, so

$$P := \|\theta^\star\|_{2,2}^2 = \sum_{i=1}^{8} \big(11\|u_i\|_2^2 + \|w_i\|_2^2\big) = 557.214136999119$$

exactly (a rational with denominator 10^{12}), and $\|\theta^\star\|_{2,2}^{12} = P^6$.

Step 2 (the margin). By Lemma 2, $h(\theta^\star)$ is the minimum over the 78 count vectors N of the smallest of the two gaps $f_N[y] - f_N[c]$, $c \neq y$. Each gap is an explicit polynomial in the weights of Table 1. The table in Appendix A lists, for every N, its label and the smaller gap rounded down to six decimals. All 78 entries are at least 2.731226; the minimum is attained at $N = (2, 0, 9)$. Hence

$$h(\theta^\star) \;\geq\; h_0 := 2.7312.$$

Step 3 (comparison). By Lemma 1,

$$\gamma^\star_{2,\mathrm{IF}} \cdot P^6 = \frac{3}{12^5 \cdot 72^6}\, P^6 = \frac{3\,(P/72)^6}{12^5} = 2.590317\ldots \;<\; 2.5904.$$

Therefore

$$\frac{h(\theta^\star)}{P^6} \;\geq\; \frac{2.7312}{2.5904}\,\gamma^\star_{2,\mathrm{IF}} \;>\; 1.054\,\gamma^\star_{2,\mathrm{IF}}.$$

Exact evaluation gives the ratio $1.054398843620\ldots$. Adding zero neurons changes neither h nor P, which gives the statement for every width $m \geq 8$. □

Remark 4 (how to check)**.** Steps 1 and 3 need only a calculator. Step 2 evaluates 156 polynomials of degree 12 at rational points. The script `exact_certificate.py` does this with Python's `fractions` module, using no floating point and no third-party code, in about 0.1 s. A float64 brute force over all $3^{11} = 177{,}147$ inputs, which does not use Lemma 2, returns the same minimum.

3 Further failures

Theorem 5 ($n = 13$)**.** *For $p = 3$, $n = 13$, $b = 2$ there is a slot-symmetric product network of width 10 (weights in `exact_certificate.py`) with $P = \|\theta\|_{2,2}^2 = 832.50414381275652$ and margin $h \geq 4.150479$. Since $\gamma^\star_{2,\mathrm{IF}} P^7 = 42\,(P/1176)^7 = 3.741993\ldots$, its normalized margin is at least $1.109\,\gamma^\star_{2,\mathrm{IF}}$ (exactly $1.109163143571\ldots$).*

Proof. The same three steps as for Theorem 3, with $\binom{15}{2} = 105$ count vectors and $\nu/2 = 7$. □

Theorem 6 ($n = 14$)**.** *For $p = 3$, $n = 14$, $b = 2$ there is a slot-symmetric product network of width 10 (family parameters in `cex_n14_family_params.json`, weights rounded to seven decimals) with exact margin $h = 3.212099\ldots$ and $P = 905.29973580696392$, whose normalized margin is at least $1.4293\,\gamma^\star_{2,\mathrm{IF}}$.*

Proof. The margin is computed exactly over the $\binom{16}{2} = 120$ count vectors. Since $\nu/2 = 15/2$ is not an integer, $P^{15/2}$ and the comparison are enclosed in interval arithmetic (`interval_certificate.py`), which gives the ratio in $[1.42934420165,\ 1.42934420166]$. □

Theorem 7 (intervals of exponents)**.** *For $p = 3$:*

1. *at $n = 12$ there is one width-9 network (weights in `cex_n12_b_interval_weights.json`) whose normalized $L_{2,b}$-margin exceeds $\gamma^\star_{b,\mathrm{IF}}$ for* every $b \in [2, 2.13]$*, by a factor of at least 1.0198;*
2. *at $n = 11$ a width-8 network of the same family as $\theta^\star$ (parameters in `cex_n11_family_params.json`, weights rounded to seven decimals) beats $\gamma^\star_{b,\mathrm{IF}}$ for every $b \in [2, 2.02]$, by a factor of at least 1.0234.*

Proof. The margin h does not depend on b and is computed exactly as above. For b in a closed interval J, the quantities $P_b = \sum_i \|\omega_i\|_2^b$, $P_b^{\nu/b}$ and $\gamma^\star_{b,\mathrm{IF}}$ are compositions of exp and log in b, so they can be enclosed by interval arithmetic. Splitting J into 260 (respectively 40) subintervals and evaluating $h/(P_b^{\nu/b}\gamma^\star_{b,\mathrm{IF}})$ in outward-rounded interval arithmetic (mpmath, 50 digits) on each subinterval gives a lower bound above 1.0198 (respectively 1.0234) on every piece. Script: `interval_certificate.py`. □

Remark 8 (the b-window grows)**.** Re-optimizing the network for each b gives at $n = 11$ ratios $1.061, 0.904, 0.852, 0.653$ at $b = 2, 2.1, 2.2, 2.5$, and at $n = 12$ ratios $1.190, 1.063, 1.007, 0.957$ at $b = 2, 2.1, 2.15, 2.2$ (the last two interval-certified as well). The best ratios found at $b = 2$ for $n = 3, \ldots, 14$ are $0.599, 0.598, 0.608, 0.695, 0.699, 0.785, 0.895, 0.895, 1.058, 1.192, 1.178, 1.429$. These are numerical lower bounds on the true maximum; they are not proved maxima.

4 What the counterexample is

Up to a common scale, every input factor in Table 1 has the form

$$u_i(a) = \kappa_i + 2\cos\bigl(2\pi a/3 + \varphi_i\bigr),$$

a single cosine plus a constant (DC) offset κ_i. The interference-free block of Proposition 9 is the case $\kappa_i = 0$ with ν neurons whose phases cover 180°. In $\theta^\star$ the phases φ_i are spaced about 14° apart on a 98° arc centred at 180°, the offsets increase from $\kappa = 0.50$ at the centre to 0.76 at the ends, and the budget is split almost equally (each neuron has $\|\omega_i\|^2/P \in [0.124, 0.127]$). The margin is attained, to within 0.2%, on 32 of the 78 count classes.

Remark 9 (why it can beat Lemma 7)**.** With $8 < \nu = 12$ neurons, Lemma 7 says that the sign-flipped copies of the useful mode cannot cancel. The constant components create the Fourier modes with $k_t = 0$ for some slot t, which Theorem 8 leaves unconstrained. These modes move the leftover interference back inside the gap cone at every input. Their price is visible through Parseval: a slot with offset κ costs $(1 + \kappa^2/2)$ times a centred slot of the same useful amplitude, here about 1.185 on average. Eight such neurons cost about 9.5 centred-neuron equivalents against twelve for the interference-free block. At $b = 2$ the normalized margin scales like (budget per neuron)$^{\nu/2}$, so the saving outweighs the loss in efficiency once ν is large enough.

5 Why own-frequency arguments cannot work

The proof of the case $p = 3, n = 2$ and the matrix proof of the interval $b \geq b_0$ both bound the margin using only (i) the clean and sign-flipped Fourier modes of the score table, and (ii) the single-neuron Fourier budget $\sum_i |\beta_{ij}|^{b/\nu} \leq B_p^{b/\nu}\|\omega_i\|_2^b$. Every mode with $k_t = 0$ for some slot is left free. We show that this information alone does not imply the conjectured bound at $p = 3$, $n = 3$, $b = 2$: a proof of any remaining small case must use the modes with a constant slot and their cost.

For $\xi \in \mathbb{C}^n$ write $\chi(\xi) \in (\mathbb{C}^2)^{\otimes n}$ for the tensor with entries $\chi(\xi)_T = \prod_{t\in T}\xi_t$, $T \subseteq [n]$. For a score table f on D_3 put $\Phi_f(x) = \sum_c f(x)[c]\,\omega^{\,c-y(x)}$ with $\omega = e^{2\pi i/3}$. Then $\min\bigl(g(x,1), g(x,2)\bigr) = \operatorname{Re}\Phi_f(x) - |\operatorname{Im}\Phi_f(x)|/\sqrt{3}$, and the own-frequency modes of f are the coefficients $M_T(f)$ of the characters $\prod_{t\in T}\omega^{x_t}$ in Φ_f. For a centred single-frequency network, $\Phi_f(x) = \sum_i \beta_i \prod_t (1 + \xi_{it}\omega^{x_t})$ with $|\xi_{it}| = 1$, so its own-frequency modes are $M = \sum_i \beta_i \chi(\xi_i)$; here β_i is 3/2 times the paper's β_{i1}. In this normalization the single-neuron budget and Lemma 1 turn the conjectured bound into $h \leq \nu^{1-s}Q^s$ with $Q = \sum_i |\beta_i|^r$, $r = 2/\nu$, $s = \nu/2$, with equality of the budget for balanced centred neurons.

Proposition 10 (barrier at $n = 3$)**.** *Let $n = 3$, $\alpha = 2\pi \cdot 0.2324$, and let $(\beta_0, \beta_+, \kappa) \in \mathbb{R} \times \mathbb{C} \times \mathbb{R}$ solve the linear system*

$$\beta_0 + 2\,\mathrm{Re}\big(\beta_+ e^{ij\alpha}\big) - 2\kappa \cos\big(\tfrac{2\pi j}{3} - \tfrac{\pi}{3}\big) = [j = 0], \qquad j = 0, 1, 2, 3.$$

Then $\beta_0 = 0.37468\ldots$, $\beta_+ = 0.44919\ldots + 0.10100\ldots i$, $\kappa = 0.27306\ldots > 0$, and with $\beta_- = \overline{\beta_+}$ and atoms $\xi_0 = (1, 1, 1)$, $\xi_\pm = e^{\pm i\alpha}(1, 1, 1)$:

1. *$\sum_{j \in \{0,\pm\}} \beta_j \chi(\xi_j) = e_\emptyset + \kappa e^{-i\pi/3} \chi(\omega \mathbf{1}) + \kappa e^{i\pi/3} \chi(\omega^2 \mathbf{1})$ exactly;*
2. *the real score table f with $\Phi_f \equiv 1$ except $\Phi_f(2, 2, 2) = 1 + 27\kappa e^{-i\pi/3}$ and $\Phi_f(1, 1, 1) = 1 + 27\kappa e^{i\pi/3}$ has margin $h(f) = 1$, and its own-frequency modes equal the tensor in (1);*
3. *$Q = |\beta_0|^{1/2} + 2|\beta_+|^{1/2} = 1.969176\ldots$, so $\nu^{1-s} Q^s = Q^2/4 = 0.969414\ldots < 1 = h(f)$.*

Proof. (1) Both sides depend on T only through $|T| = j$. The j-th entry of the left side is $\beta_0 + \beta_+ e^{ij\alpha} + \overline{\beta_+} e^{-ij\alpha}$, and that of the right side is $[j = 0] + 2\kappa \cos(2\pi j/3 - \pi/3)$. These agree by the defining system. The numerical values follow by solving the 4×4 system. (2) Write $\Phi = 1 + \Delta$ with Δ supported on two points. The coefficient of $\prod_{t \in T} \omega^{x_t}$ in a point mass δ_x is $3^{-3} \prod_{t \in T} \omega^{-x_t}$, which is $3^{-3} \chi(\omega^{-x})_T$. Since $\omega^{-(2,2,2)} = \omega \mathbf{1}$ and $\omega^{-(1,1,1)} = \omega^2 \mathbf{1}$, the own-frequency modes of Φ_f are those in (1). A real table with prescribed Φ exists: set $f(x)[y(x)] = 0$ and $f(x)[y(x) - \delta] = -g_\delta(x)$, where $\Phi(x) = g_1(x) e^{i\pi/3} + g_2(x) e^{-i\pi/3}$. Because $27\kappa e^{\mp i\pi/3}$ lies in the cone generated by $e^{\pm i\pi/3}$ and $1 = e^{i\pi/3} + e^{-i\pi/3}$, all gaps are at least 1, and the minimum 1 is attained. (3) Direct evaluation. □

So the own-frequency data of a table with margin 1 is matched by a centred three-neuron network whose budget would allow only margin 0.9694 under the conjectured bound. Equivalently, the relaxation exceeds the bound by the factor 1.0316 (at the optimal α: 1.031555). At $n = 4$ the same construction gives 1.708 (`barrier_certificate.py`). Numerically, the chain-matrix relaxation with the gap cone admits ratios up to about 1.245 at $n = 3$, $b = 2$, and about 1.034 even when restricted to networks without DC components.

6 Status and corrected questions

- **Unaffected.** Theorem 8, Proposition 9, Corollary 14 ($b = \nu$), Theorems 10–11 and Corollary 12 (orders in p at fixed ν, b), the full proof at $p = 3, n = 2$, and the interval $b \in [b_0, \nu]$.
- **False.** Conjecture 20 as stated: certified at $p = 3$, $b = 2$ for $n = 11, 13, 14$, and for every $b \in [2, 2.13]$ at $n = 12$ (Theorems 3–7).
- **Open.** At $b = 2$, searches over all real product networks found nothing above $\gamma^\star_{b,\mathrm{IF}}$ for $p = 3$, $n \le 6$ (and the slot-symmetric family stays below it for $n \le 10$), for $p = 5$, $n \le 3$, and for $p = 7$, $n = 2$. The boxed case $h \le P^2/768$ ($p = 3, n = 3, b = 2$) is still plausible. By Proposition 10, proving it requires charging the cost of constant components.

Question 11. For each p, n let $\mathcal{B}(p, n) \subseteq [2, \nu]$ be the set of exponents at which $\gamma^\star_{b,\mathrm{IF}}$ is the true maximum. Is $\mathcal{B}(p, n)$ an interval $[b^\star(p, n), \nu]$? We know $b^\star(p, n) \le b_0(p, n)$, $b^\star(3, 2) = 2$, $b^\star(3, 11) > 2.02$ and $b^\star(3, 12) > 2.13$. Is $b^\star(3, n) = 2$ for $n \le 10$, and does $b^\star(3, n) \to \infty$, or does the ratio at $b = 2$ grow without bound as $n \to \infty$?

A The 78 gaps of $\theta^\star$

For each count vector $N = (N_0, N_1, N_2)$, $|N| = 11$: the label $y \equiv N_1 + 2N_2$ and $\min_{c \neq y}(f_N[y] - f_N[c])$, rounded down to six decimals (exact rational evaluation).

N	y	gap	N	y	gap	N	y	gap
(0,0,11)	1	3.051329	(2,3,6)	0	2.742391	(5,2,4)	1	2.771488
(0,1,10)	0	2.803939	(2,4,5)	2	4.031556	(5,3,3)	0	2.759766
(0,2,9)	2	60.206995	(2,5,4)	1	4.005469	(5,4,2)	2	2.761313
(0,3,8)	1	2.740909	(2,6,3)	0	2.742762	(5,5,1)	1	7.990881
(0,4,7)	0	2.742387	(2,7,2)	2	2.740830	(5,6,0)	0	2.800768
(0,5,6)	2	4.007730	(2,8,1)	1	58.780606	(6,0,5)	1	2.740790
(0,6,5)	1	3.988606	(2,9,0)	0	2.978258	(6,1,4)	0	2.742287
(0,7,4)	0	2.742568	(3,0,8)	1	2.739929	(6,2,3)	2	4.078604
(0,8,3)	2	2.740642	(3,1,7)	0	2.741317	(6,3,2)	1	4.039402
(0,9,2)	1	58.541693	(3,2,6)	2	8.073045	(6,4,1)	0	2.743053
(0,10,1)	0	2.739688	(3,3,5)	1	2.756772	(6,5,0)	2	2.741107
(0,11,0)	2	2.998367	(3,4,4)	0	2.750082	(7,0,4)	2	8.094945
(1,0,10)	2	3369.126003	(3,5,3)	2	2.751623	(7,1,3)	1	2.786274
(1,1,9)	1	2.740041	(3,6,2)	1	7.982238	(7,2,2)	0	2.769483
(1,2,8)	0	2.741426	(3,7,1)	0	2.770576	(7,3,1)	2	2.771036
(1,3,7)	2	8.062049	(3,8,0)	2	2.770171	(7,4,0)	1	7.999486
(1,4,6)	1	2.742128	(4,0,7)	2	60.569757	(8,0,3)	0	2.742177
(1,5,5)	0	2.740313	(4,1,6)	1	2.740868	(8,1,2)	2	4.100176
(1,6,4)	2	2.741967	(4,2,5)	0	2.742358	(8,2,1)	1	4.054570
(1,7,3)	1	7.973557	(4,3,4)	2	4.055505	(8,3,0)	0	2.743148
(1,8,2)	0	2.740562	(4,4,3)	1	4.022401	(9,0,2)	1	2.801133
(1,9,1)	2	2.740160	(4,5,2)	0	2.742924	(9,1,1)	0	2.779233
(1,10,0)	1	3219.508645	(4,6,1)	2	2.740985	(9,2,0)	2	2.780792
(2,0,9)	0	2.731226	(4,7,0)	1	59.020476	(10,0,1)	2	4.121851
(2,1,8)	2	60.388104	(5,0,6)	0	2.741182	(10,1,0)	1	4.069074
(2,2,7)	1	2.740907	(5,1,5)	2	8.084010	(11,0,0)	0	2.789017